3. *David G. Ullman, “The Mechanical Design Process”, Sixth Edition, David Ullman LLC.*
4. *Benjamin W Nishel and Alan B Draker, “Product Design & Process Engineering”, McGraw Hill.*

Reference Books

1. *Saaksvuori Antti, Milmmonen Ansel, “ Product Life Cycle Management” Springer.*
2. *John Stark, “Product Lifecycle Management: Paradigm for 21st Century Product Realisation”, Springer-Verlag.*
3. *Burden, Rodger, “PDM: Product Data Management”, Resource Pub.*
4. *Jerry Clement, Andy Coldrick and John Sari, “Manufacturing Data Structures”, John Wiley Sons.*

Evaluation Scheme:

S. No.	**Evaluation Elements**	**Weightage (%)**
1.	MST	35
2.	EST	35
3.	Sessional (Assignments/Projects/Tutorials/Quizzes)	30

UME747 : Processing of Polymers And Composites

L	T	P	Cr
3	0	0	3.0

Course Objective: To impart knowledge of the basic nature of different polymers and manufacturing processes associated thereof. Tailoring properties in composites as required for specific applications. To introduce attendants to the principles of the processing and concept of the deformation behaviour of plastics. To provide an outline account for all major processing routes, thermoplastics, as well as thermoset and rubbers.

Syllabus

Properties and processing of polymers: Structure and mechanical properties of plastics: thermoplastics and thermosets, their properties and applications, processing the polymers considering crosslinking and curing, influence of time, temperature, and mass, shelf life and pot life, stoichiometric considerations. additives in polymers: dispersion aids, UV stabilizers, antioxidants and antiozonents, processing/flow modifiers, different fillers.

Extrusion using single and twin screw extruders, injection moulding, thermoforming, compression moulding, transfer moulding, general behavior of polymer melts, machining of polymers, processing of rubbers, testing of polymers, Recycling of plastics.

Properties and processing of composites: Classification of composite materials, properties of composites, processing methods of polymeric matrix composites: Hand lay-up, autoclaving, filament winding, pultrusion, compression molding, pre-pegging, sheet molding compounds etc.,

Secondary processing of composite materials, need of secondary operations, different type of secondary operations, machining, drilling, joining of composites. welding of polymers using thermal, ultrasonic and laser bonding, destructive and non-destructive4and evaluation, characterization of composites using microscopy: Scanning electron microscopy and transmission electron microscopy, review of simulation technologies for composite design, manufacturing and performance, applications of polymer composites in automotive, marine and aerospace

Research Assignment: Students in a group of 4/5 will do term projects with help of critically reviewing some technical research papers on the recent technology developments and industrial applications as well as challenges for processing of polymers and composites.

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze the behavior of polymers, their properties to select suitability for engineering applications.
2. Know the behavior during processing of polymers.
3. Gain knowledge on the properties and industrial applications of the polymers.
4. Gain practical knowledge of the structure-property relationships to improve properties of polymers and the manufacturing the products with alternative technology.
5. Derive and calculate stress, strain and modulus for a given problem of unidirectional composite.

Text Books